

Testing Portfolio Models on the Sectors of the S&P 500

Miura Financial Research
*Q*uantitative Research Team

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The members who contributed to this work are:
Giuseppe Zuco - Head of Quantitative Research Team
Francesco De Simone - Head of Quantitative Research Team
Lorenzo Benzakour - Quant Analyst
Irene Balan - Quant Analyst
Cristian Losi - Quant Analyst
Simone D' Isabella - Quant Analyst
Gianluca De Pieri - Quant Analyst
Francesco Melocchi - Quant Analyst
Alberto Preti - Quant Analyst
Ernesto Ruschena - Quant Analyst
Luca Girlando - Quant Analyst
Tommaso Rossini - Quant Analyst
Angelo Di Martino - Quant Analyst

All the members contributed equally to this work

Introduction

In finance and economic theory, portfolio models and optimization serve as the foundation for both academic research and practical asset management. A central tenet is the mean–variance framework developed by Markowitz [11], which formalizes the trade-off between expected return and risk by using variance as the key measure of uncertainty. This paradigm has profoundly influenced portfolio choice and asset pricing, as evidenced by its role in the development of the Capital Asset Pricing Model (CAPM) introduced by Sharpe [18]. However, the elegance of mean–variance analysis is tempered by its reliance on several strong assumptions—such as the normality of returns, quadratic utility functions, and frictionless markets—that may not fully capture the complexities of actual financial systems. Scholar such as Merton [12], has emphasized the limitations of its uncritical application, highlighting the challenges posed by real-world phenomena like fat-tailed distributions, skewness, and serial correlations.

Building on these theoretical considerations, our empirical investigation seeks to test whether different portfolio construction methodologies can effectively outperform the S&P 500 benchmark. To this end, we employ exchange-traded funds (ETFs) provided by SPDR that track the S&P 500 index. Specifically, we assess traditional approaches such as equally weighted, mean–variance, risk parity, and volatility parity portfolios, alongside a novel method known as Hierarchical Risk Parity, as proposed by De Prado [10]. In order to rigorously examine the sensitivity of portfolio performance to estimation and rebalancing parameters, we adopt a rolling window approach in which portfolios are constructed over estimation horizons of 12 and 18 months and rebalanced at frequencies of 1 and 3 months. This methodology results in four distinct configurations that allow for a detailed exploration of their relative performance.

The constructed portfolios are evaluated using an extensive set of risk-adjusted performance measures. This evaluation encompasses established metrics such as the Sharpe ratio (Sharpe, [17]) and Jensen's alpha (Jensen, [7]), alongside additional measures like the Sortino ratio and a probabilistic variant of the Sharpe ratio (see Bailey and de Prado [1]). Furthermore, market timing models, including the Treynor and Mazuy model [15] and the Henriksson and Merton model [6], are also employed.

1 S&P 500 Sectors

Select Sector SPDRs S&P 500

The data used in this analysis are the sectors ETF from SPDR[1]. These data were used, mainly to observe if investing in these sectorial ETF might generate some advantage with respect to investing in the S&P 500 itself.

More specifically, the Select Sector SPDR ETFs each track a distinct sector of the S&P 500 Index, using underlying indices that segment S&P 500 companies by their Global Industry Classification Standard (GICS) sector. Each fund's index is composed exclusively of S&P 500 constituent companies belonging to one of the eleven GICS sectors (e.g. Technology, Health Care, etc.). These indices are market capitalization-weighted with a proprietary modified weighting methodology that caps large individual stock weights to meet U.S. Internal Revenue Code diversification requirements. The index provider, S&P Dow Jones Indices (S&P DJI), assigns each S&P 500 stock to the appropriate sector index based on its GICS industry classification, and maintains the indices in tandem with S&P 500 changes (there is no fixed annual reconstitution – when a company is added to or removed from the S&P 500, or its sector classification changes, the sector indices are adjusted accordingly). S&P DJI serves as the index calculation agent for these benchmarks. Each Select Sector SPDR Fund in turn aims to replicate the performance of its sector index by holding essentially all the index's component stocks in the same weightings (under normal conditions at least 95% of each fund's assets are invested in securities of its index). Below is a brief description of the construction of each sector index and its corresponding ETF, for more details see the prospectus[1].

Communication Services (XLC) Index Composition All S&P 500 companies classified by GICS as Communication Services firms are included. This covers industries such as telecommunication services, media, entertainment, and interactive media & services. Index methodology: the index is market-cap weighted with modified weights to limit any single company's concentration (ensuring compliance with tax diversification rules). It is one of the eleven S&P 500 sector indices maintained by S&P DJI, with membership strictly drawn from S&P 500 constituents. ETF Tracking Strategy: the XLC fund uses a full replication approach, holding substantially all the index's component stocks in approximately their index weights.

Consumer Staples (XLP) Index Composition All S&P 500 companies classified by GICS as Consumer Staples are included. This encompasses industries such as food & beverage, household and personal products, tobacco, and staples retailing. Index Methodology: like the others, it is a float-adjusted market cap index with modified weighting to prevent any single Consumer Staples stock from exceeding the concentration limits. The index is derived entirely from S&P 500 constituents in the Consumer Staples sector. ETF Tracking Strategy: the XLP fund fully replicates the index by investing

[1]Select Sector SPDRs are unique Exchange Traded Funds that unbundle the benchmark S&P 500 and give investors ownership in particular sectors or groups of industries that are components of the S&P 500 Index. The eleven Funds together comprise all of the companies included in the S&P 500. For more detail see the website: SPDR,

in virtually all the constituent stocks in their index proportions, maintaining a high correlation with the index.

Energy (XLE) Index Composition Comprises all S&P 500 companies classified by GICS as Energy sector firms, including those in the oil, gas & consumable fuels industry and energy equipment & services industry. Index Methodology: the index is market-cap weighted with modifications to limit concentration in any one Energy company. It is one of the S&P 500 sector indices, so only S&P 500 members in the Energy sector are included. ETF Tracking Strategy: the XLE fund uses a replication strategy, holding the full range of Energy sector stocks from the index in proportionate weights. This approach closely aligns the fund's performance with the underlying index

Financials (XLF) Index Composition Includes all S&P 500 constituents classified by GICS as Financials. This spans banks, insurance, financial services, capital markets companies, consumer finance, and even financial REITs (mortgage REITs) within the S&P 500. Index Methodology: it is a modified market-cap weighted index of S&P 500 Financial sector stocks. The largest financial companies' weights are adjusted as needed to meet the index's single-stock cap rules. Only S&P 500 members are selected. ETF Tracking Strategy: the XLF fund holds essentially all the index constituents, aiming for full replication. Under normal conditions, at least 95% of the portfolio is invested in the index's component securities, which facilitates tracking the index closely

Health Care (XLV) Index Composition All S&P 500 companies classified by GICS as Health Care are included. These include industries such as pharmaceuticals, health care equipment & supplies, health care providers & services, biotechnology, life sciences tools & services, and health technology. Index Methodology: the index is market-cap weighted with modifications to cap concentrations, following the same proprietary methodology applied to all Select Sector indices. Only S&P 500 constituent stocks in the Health Care sector form this index. ETF Tracking Strategy: XLV uses a replication strategy, investing in the vast majority of the index's healthcare stocks in their respective weights. This ensures the fund's performance corresponds closely to the Health Care sector index

Industrials (XLI) Index Composition Covers all S&P 500 companies in the GICS Industrials sector. It spans a broad range of industrial industries, including aerospace & defense, industrial conglomerates, machinery, transportation, construction & engineering, electrical equipment, professional services, and commercial services & supplies. Index Methodology: the index is weighted by market capitalization (float-adjusted) with caps on dominant stocks to adhere to diversification rules. Only S&P 500 members classified as Industrials are included. ETF Tracking Strategy: XLI holds the full set of index constituents in their indexed weights wherever practical. By replicating the index's composition, the fund delivers performance in line with the S&P 500 Industrials sector.

Materials (XLB) Index Composition All S&P 500 companies classified as Materials by GICS are included. This comprises industries such as chemicals, metals & mining, paper & forest products, containers & packaging, and construction materials. Index Methodology: the Materials sector index is market-capitalization weighted with single-stock weight limits applied per the standard Select Sector

index methodology. It includes only S&P 500 constituent companies in the Materials sector. ETF Tracking Strategy: the XLB fund fully replicates the index by holding substantially all the Materials sector stocks from the S&P 500 in appropriate proportions. This replication approach yields performance tightly aligned with the underlying index.

Real Estate (XLRE) Index Composition Composed of all S&P 500 companies in the GICS Real Estate sector. It includes Real Estate Investment Trusts (REITs) and real estate management & development firms. Index Methodology: the index is float-adjusted market cap weighted with modifications to limit large concentrations. Only S&P 500 constituents classified as Real Estate are included. ETF Tracking Strategy: XLRE employs a replication strategy, investing in essentially all the S&P 500 Real Estate sector components in the index's weightings. This allows the fund to closely mirror the Real Estate sector index's returns.

Technology (XLK) Index Composition Encompasses all S&P 500 companies classified by GICS as Information Technology. This includes technology hardware, storage & peripherals, software, communications equipment, semiconductors & semiconductor equipment, IT services, and electronic equipment. Index Methodology: the index is market-cap weighted with capped components where necessary to prevent any single tech stock from dominating. It is limited to S&P 500 constituent stocks in the Information Technology sector. ETF Tracking Strategy: The XLK fund holds virtually all the index constituents, replicating the index. It normally invests the vast majority of its assets in the S&P 500 Information Technology sector stocks that make up the index, thereby closely tracking the index's performance.

Utilities (XLU) Index Composition Consists of all S&P 500 companies in the GICS Utilities sector. It includes electric, gas and water utilities, multi-utilities, as well as independent power producers & energy traders. Index Methodology: the index follows the standard Select Sector modified market cap weighting, capping any outsized utility company weights to meet diversification thresholds. All constituents are drawn from the S&P 500 Utilities sector. ETF Tracking Strategy: the XLU fund fully replicates the Utilities index by holding the S&P 500 utilities sector stocks in proportion to their index weights. This replication approach ensures the fund's performance reflects the underlying Utilities sector index. Each of the above sector indices is maintained by S&P Dow Jones Indices with the same general methodology each index comprises the S&P 500 members of one GICS sector, weighted by market cap with limits on single-stock exposure.

To conclude, these indices have no separate rebalancing schedule; they update automatically with changes in S&P 500 membership or GICS classifications. The Select Sector SPDR ETFs that track these indices use a replication investment strategy to closely match the index composition and returns, providing investors targeted exposure to the respective S&P 500 sector while maintaining the broad characteristics of the S&P 500's construction. Additional Index Information: the sector indices are compiled by assigning each S&P 500 stock to one of the Select Sector indices according to its GICS sector. Because the S&P 500 itself is maintained continuously, the sector indices likewise change whenever the S&P 500's composition changes. The weighting scheme of the Select Sector indices imposes

caps on the largest constituents – this "modified market capitalization" approach means the sector index weights may differ from S&P's uncapped sector indices. This modification is done to conform to the Internal Revenue Code's requirements for fund diversification, ensuring no single stock dominates a sector index beyond allowable thresholds. S&P DJI acts as the calculation agent for the indices and publishes their values but does not sponsor or endorse the ETFs themselves. The creation of each sector index is a rules-based process grounded in the GICS classification of S&P 500 constituents, with transparent weighting adjustments for concentration limits, enabling the corresponding ETF to effectively capture the performance of that sector in a manner consistent with the overall S&P 500 index structure.

2 Portfolio Models

In this section, we present a concise theoretical overview of the models considered in this analysis. The portfolio models examined include: equally weighted, mean–variance, minimum variance, risk parity, volatility parity, and hierarchical risk parity. These brief explanations provide the foundation for understanding the rationale underlying the use of these models.

Equally Weighted Portfolio

An equally weighted portfolio (EW) is a common strategy used to diversify across different assets. This approach does not involve any optimization or estimation and completely disregards the data. It assigns equal importance to each asset in a portfolio, index, or index fund, without considering the size of the companies. As a result, stocks from smaller companies are given the same weight, or statistical significance, as those from larger companies when evaluating overall performance.

It is possible to formalize this approach with the following notation:

$$\mathbf{w} \quad \text{s.t.} \quad w_i = w_j \quad \sum_{i=1}^{n} w_i = 1 \quad w_i \geq 0 \quad \forall i, j = 1, \ldots, n. \tag{1}$$

In Equation (1), we assume that the weight of asset i, denoted w_i, is equal to the weight of asset j, with the constraint that their sum equals one. This notation is then generalized to accommodate n assets.

Equally weighted portfolios, also referred to as $1/N$ portfolios, are commonly used in practice, as demonstrated by Benartzi and Thaler [2], and Windcliff and Boyle [16].These portfolios have been demonstrated to be efficient in out-of-sample analyses.

Despite its simplicity, the equally weighted portfolio raises questions about the utility of more sophisticated portfolio models. In fact, the study by De Miguel, Garlappi, and Uppal [4] found that none of the optimization models proposed in the literature consistently achieve a higher Sharpe ratio or CEQ (certainty equivalent) return than the $1/N$ portfolio. Additionally, the $1/N$ portfolio benefits from very low turnover.

Moreover, if all assets have the same correlation coefficient, as well as identical means and variances, the equally weighted portfolio becomes the unique portfolio on the efficient frontier.

The concept of equally weighted portfolios has gained increasing attention, fueled by the historical performance of small-cap stocks and the growth of numerous exchange-traded funds (ETFs). In addition, we recall, that the actual need of equally weight portfolios, is demonstrated by the presence of some Standard & Poor's equally weighted indexes[2].

Mean-Variance optimization and GMV Portfolios

The Mean-Variance (MV) optimization framework, introduced by Harry Markowitz in 1952, [11], serves as the cornerstone of Modern Portfolio Theory. The core principle underlying this approach is

[2]Equal-weight indices include the same constituents as their respective market-cap-weighted indices, but each company is allocated a fixed equal weight in the index at each quarterly rebalance. See S&P Global for details.

the reduction of portfolio variance through diversification in the mean-variance sense, by allocating across assets whose returns exhibit low covariances. This methodology entails determining a weight vector, denoted by $\mathbf{w}$, that minimizes the portfolio volatility, $\sigma_P(\mathbf{w})$, for a given level of expected return $\mu_P(\mathbf{w})$. Let $\mathbf{w}$ be a column vector of dimension n, composed of elements $\mathbf{w} = (w_1, w_2, \ldots, w_n)$, representing the proportions invested in each asset. The covariance matrix Σ is a symmetric, positive definite, and non-singular $n \times n$ matrix containing the pairwise covariances between asset returns. Non-singularity of Σ implies that no asset is perfectly correlated with a portfolio composed of the remaining assets, and that none of the assets (or portfolios) is riskless. The expected return of the portfolio is given by

$$R_P = \mathbf{w}^\top \boldsymbol{\mu},$$

where $\boldsymbol{\mu}$ is the vector of expected asset returns. The variance of the portfolio is expressed as:

$$\sigma_P^2(\mathbf{w}) = \mathbf{w}^\top \Sigma \mathbf{w}. \tag{2}$$

Thus, the MV optimization problem can be formulated as the following quadratic program:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \mathbf{w}^\top \Sigma \ \mathbf{w} \\ \text{s.t.} \quad & \mathbf{w}^\top \boldsymbol{\mu} = \mu_p \\ & \mathbf{1}^\top \mathbf{w} = 1 \end{aligned}$$

The purpose is to minimize a convex function (2), subject to two linear constraints. The objective function is convex due to the positive definiteness of Σ, and the constraint set is convex as well. Consequently, the optimization problem admits a unique solution, which can be derived by solving the first-order conditions (FOCs) of the associated Lagrangian. The Lagrangian for this problem is constructed as follows:

$$\mathcal{L}(\mathbf{w}, \lambda, \gamma) = \mathbf{w}^\top \Sigma \mathbf{w} - \lambda \left(\mathbf{w}^\top \boldsymbol{\mu} - \mu_p \right) - \gamma \left(\mathbf{1}^\top \mathbf{w} - 1 \right).$$

Taking partial derivatives with respect to $\mathbf{w}$, λ, and γ yields the FOCs:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \lambda} &= -\left(\mathbf{w}^\top \boldsymbol{\mu} - \mu_p \right) = 0 \quad &\Rightarrow \quad \mathbf{w}^\top \boldsymbol{\mu} = \mu_p \\ \frac{\partial \mathcal{L}}{\partial \gamma} &= -\left(\mathbf{1}^\top \mathbf{w} - 1 \right) = 0 \quad &\Rightarrow \quad \mathbf{1}^\top \mathbf{w} = 1 \\ \frac{\partial \mathcal{L}}{\partial \mathbf{w}} &= 2\Sigma \mathbf{w} - \lambda \boldsymbol{\mu} - \gamma \mathbf{1} = 0 \quad &\Rightarrow \quad 2\Sigma \mathbf{w} = \lambda \boldsymbol{\mu} + \gamma \mathbf{1}, \end{aligned}$$

where $\mathbf{0}$ is an n-dimensional vector of zeros. Solving the first-order conditions leads to the expression for the optimal weight vector:

$$\mathbf{w}^* = \frac{1}{2} \Sigma^{-1} (\lambda \boldsymbol{\mu} + \gamma \mathbf{1}) = \frac{1}{2} \Sigma^{-1} \begin{bmatrix} \boldsymbol{\mu} & \mathbf{1} \end{bmatrix} \begin{bmatrix} \lambda \\ \gamma \end{bmatrix}. \tag{3}$$

To enforce the linear constraints in a compact matrix notation:

$$\begin{cases} \mathbf{w}^\top \boldsymbol{\mu} = \mu_p \\ \mathbf{w}^\top \mathbf{1} = 1 \end{cases} \quad \Leftrightarrow \quad \mathbf{w}^\top \begin{bmatrix} \boldsymbol{\mu} & \mathbf{1} \end{bmatrix} = \begin{bmatrix} \mu_p & 1 \end{bmatrix}.$$

Substituting the expression for (3) into the constraints and simplifying yields:

$$\frac{1}{2}\left(\begin{bmatrix}\boldsymbol{\mu} & \mathbf{1}\end{bmatrix}^\top \Sigma^{-1} \begin{bmatrix}\boldsymbol{\mu} & \mathbf{1}\end{bmatrix}\right)\begin{bmatrix}\lambda \\ \gamma\end{bmatrix} = \begin{bmatrix}\mu_p \\ 1\end{bmatrix}.$$

Let A denote the resulting 2×2 symmetric and positive definite matrix, with entries:

$$A = \begin{bmatrix}a & b \\ c & d\end{bmatrix}$$

where:

$$a = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}, \quad b = c = \boldsymbol{\mu}^\top \Sigma^{-1} \mathbf{1}, \quad d = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}.$$

Provided that at least one of the Lagrange multipliers is non-zero, we obtain:

$$\frac{1}{2} A \begin{bmatrix}\lambda \\ \gamma\end{bmatrix} = \begin{bmatrix}\mu_p \\ 1\end{bmatrix}.$$

Since A is nonsingular and its inverse exists, it is possible to solve for the multipliers:

$$\frac{1}{2}\begin{bmatrix}\lambda \\ \gamma\end{bmatrix} = A^{-1}\begin{bmatrix}\mu_p \\ 1\end{bmatrix}. \tag{4}$$

Substituting (4) into (3) leads to a final expression for $\mathbf{w}^*$:

$$\mathbf{w}^* = \frac{1}{2}\Sigma^{-1}\begin{bmatrix}\boldsymbol{\mu} & \mathbf{1}\end{bmatrix}\begin{bmatrix}\lambda \\ \gamma\end{bmatrix} = \Sigma^{-1}\begin{bmatrix}\boldsymbol{\mu} & \mathbf{1}\end{bmatrix} A^{-1}\begin{bmatrix}\mu_p \\ 1\end{bmatrix}.$$

Given that Σ is positive definite and that $\boldsymbol{\mu}$ does not consist of identical values, this solution is unique and characterizes the optimal weights for the minimum variance portfolio targeting expected return μ_p. To evaluate the variance of any minimum variance portfolio for a given μ_p, we compute:

$$\sigma^2_{\mathbf{w}^*} = (\mathbf{w}^*)^\top \Sigma (\mathbf{w}^*) = \begin{bmatrix}\mu_p & 1\end{bmatrix} A^{-1}\left(\begin{bmatrix}\boldsymbol{\mu} & \mathbf{1}\end{bmatrix}^\top \Sigma^{-1}\Sigma\Sigma^{-1}\begin{bmatrix}\boldsymbol{\mu} & \mathbf{1}\end{bmatrix}\right) A^{-1}\begin{bmatrix}\mu_p \\ 1\end{bmatrix}.$$

Simplifying:

$$\sigma^2_{\mathbf{w}^*} = \begin{bmatrix}\mu_p & 1\end{bmatrix} A^{-1}\begin{bmatrix}\mu_p \\ 1\end{bmatrix}.$$

Recalling that $A = \begin{bmatrix}a & b \\ c & d\end{bmatrix}$, with determinant $ad - c^2 \neq 0$ (where $b = c$), the inverse of A is given by:

$$A^{-1} = \frac{1}{ad - c^2}\begin{bmatrix}d & -c \\ -c & a\end{bmatrix}.$$

Thus, it is possible to rewrite $\sigma^2_{\mathbf{w}^*}$ as:

$$\sigma^2_{\mathbf{w}^*} = \frac{1}{ad - c^2}\begin{bmatrix}\mu_p & 1\end{bmatrix}\begin{bmatrix}d & -c \\ -c & a\end{bmatrix}\begin{bmatrix}\mu_p \\ 1\end{bmatrix}.$$

Expanding:

$$= \frac{1}{ad - c^2} \left(\mu_p^2 d - \mu_p c - \mu_p c + a \right)$$

$$= \frac{\mu_p^2 d - 2\mu_p c + a}{ad - c^2}.$$

This results in a parabolic relationship between portfolio variance and expected return, known as the minimum variance frontier. The upper branch of this frontier represents the set of mean-variance efficient portfolios — portfolios that offer the lowest possible risk for a given expected return. In contrast, portfolios located on the lower segment of the parabola are deemed inefficient, as they provide suboptimal risk-return trade-offs. The following Figure provide this result.

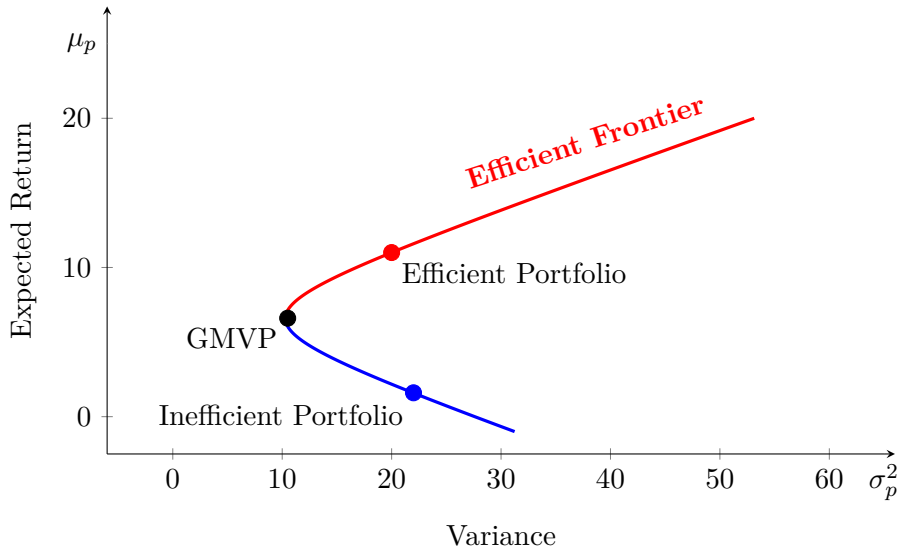


Figure 1: *Portfolio frontier in the mean-variance plane.*

In Figure 1, we observe the presence of the Global Minimum Variance Portfolio (GMVP), which lies at the vertex of this frontier and corresponds to the portfolio with the absolute lowest variance across all feasible expected returns. The expected return of the GMVP is given by:

$$R_G = \frac{b}{c}$$

and its variance is:

$$\sigma_G^2 = \frac{1}{c}.$$

By substituting R_G for μ_p in the general expression for $\mathbf{w}^*$, the weight vector for the GMVP simplifies to:

$$\mathbf{w}_G = \Sigma^{-1} \begin{bmatrix} \boldsymbol{\mu} & \mathbf{1} \end{bmatrix} A^{-1} \begin{bmatrix} R_G \\ 1 \end{bmatrix} = \frac{\Sigma^{-1}\mathbf{1}}{c}.$$

Risk Parity Portfolio

Risk parity is an asset allocation strategy that aims to distribute risk equally across portfolio components, as opposed to traditional approaches such as mean-variance optimization or equally

weighted portfolios. The concept of risk parity ensures that each asset contributes the same level of risk to the total portfolio, leading to diversification benefits and improved robustness in various market conditions. Unlike mean-variance optimization, which depends on expected returns, risk parity relies solely on volatility and the covariance structure, making it less sensitive to estimation errors. Risk parity has been widely adopted in multi-strategy hedge funds and institutional investment frameworks, as it offers an alternative to minimum-variance portfolios while maintaining better diversification. The fundamental idea is to equalize the risk contribution of each asset rather than merely equalizing portfolio weights. By doing so, risk parity seeks to mitigate concentration risk and enhance long-term stability.

Let a portfolio be represented as $\mathbf{w} = (w_1, \ldots, w_n)$, where each w, $i = 1, \ldots, n$, denotes the weight of asset in the portfolio. The total portfolio risk, represented by its volatility, is given by:

$$\sigma(\mathbf{w}) = \sqrt{\mathbf{w}^T \Sigma \mathbf{w}}. \tag{5}$$

Where, in (5), Σ is the variance covariance matrix, and $\mathbf{w}$, is the vector of portfolio weights. We define then, the marginal risk contribution of asset i as follows:

$$\partial_{w_i}(\sigma) = \frac{w_i \sigma_i^2 + \sum_{i \neq j} w_i \sigma_{ij}}{\sigma(w)}$$

measures the sensitivity of the portfolio's risk to small changes in the weights of components, σ_i^2 is the variance of the asse i, whereas, σ_{ij} is the covariance between asset i and j. . The total risk contribution of asset i is then defined as:

$$\mathrm{RC}_i = w_i \partial_{w_i}(\sigma).$$

Risk parity is achieved when all assets contribute equally to the portfolio's total risk, i.e.,

$$\mathrm{RC}_1 = \mathrm{RC}_2 = \cdots = \mathrm{RC}_n.$$

This ensures that no single asset dominates the risk exposure of the portfolio. Analytically, the problem can be formalized as:

$$\mathbf{w}^* = \left\{ \mathbf{w} \in [0,1]^n : \sum_{i=1}^{N} w_i = 1, \quad w_i \times \partial_{w_i} \sigma(\mathbf{w}) = w_j \times \partial_{w_j} \sigma(\mathbf{w}) \quad \forall \quad i, j \right\}$$

Risk parity portfolios are commonly compared to equally weighted $(1/N)$ portfolios and minimum-variance portfolios. Equally weighted $(1/N)$ portfolios assign equal weights to all assets without considering risk contributions, which can lead to concentration risk, especially when the assets have heterogeneous volatilities. On the other hand, minimum-variance portfolios aim to minimize the overall portfolio variance, but this approach often results in concentration in low-volatility assets, thus reducing the diversification benefits. In contrast, risk parity portfolios balance the risk contributions of each asset, typically leading to better diversification and greater robustness against parameter estimation errors. Mathematically, risk parity solutions lie between equally weighted and minimum-variance portfolios.

$$\sigma_{\mathrm{MV}} \leq \sigma_{\mathrm{RP}} \leq \sigma_{\mathrm{EW}}.$$

Where σ_{MV} denotes the volatility of the minimum variance portfolio, and σ_{EW} represents the volatility of the equally weighted portfolio.

Despite its advantages, the risk parity approach presents several challenges and limitations. One key issue is computational complexity, as solving nonlinear optimization problems makes the implementation of risk parity more demanding compared to simpler portfolio allocation methods. Additionally, leverage sensitivity poses a risk, since many risk parity portfolios use leverage to equalize risk contributions, which can magnify losses during market downturns. The assumption of stable correlations among assets may not always hold, leading to dynamic shifts in correlations that can cause unexpected changes in portfolio risk exposure. Furthermore, the need for frequent rebalancing to maintain risk parity increases turnover and transaction costs, which may erode the diversification benefits. Lastly, the approach is highly sensitive to volatility estimation, relying on precise calculations of asset volatilities and covariances, which can be difficult to achieve in volatile market conditions.

Understanding these limitations is crucial for investors contemplating risk parity as a strategic allocation method. Future research could aim to refine optimization techniques, integrate alternative risk measures, and explore dynamic adjustments to correlation structures.

Volatility Parity Portfolio

Volatility Parity (VP), also referred to as Naive Risk Parity, emerges as a specific case within the broader Risk Parity (RP) framework. Under certain conditions, VP represents a common ground between Mean-Variance optimization (MV) and Risk Parity (RP). Given the substantial differences between these two allocation methodologies, a direct comparison is generally not feasible. These conditions arise when the assets in the portfolio exhibit uniform pairwise correlations. In this scenario, the covariance matrix conveys no additional information beyond its diagonal elements. As a result, the RP allocation converges to VP. Similarly, though less intuitively, in the context of MV optimization, VP emerges when all pairwise correlations are identical, and all assets in the portfolio have a similar excess return per unit of risk—in other words, have similar Sharpe Ratios. The core intuition behind VP is to equalize the risk contribution of individual assets in the portfolio by underweighting riskier assets and overweighting less risky ones.

Using volatility as the risk measure, the portfolio weights are assigned in proportion to the inverse of each asset's risk. In a fully invested portfolio, the weight of each asset i is given by:

$$\mathbf{w} \quad \text{s.t.} \quad w_{\text{VP},i} = \frac{\sigma_i^{-1}}{\sum_{j=1}^{n} \sigma_j} \quad \sum_{i=1}^{n} w_{\text{VP},i} = 1 \quad w_{\text{VP},i} \geq 0 \quad \forall i, j = 1, \ldots, n.$$

In matrix notation, we would have:

$$\mathbf{w}_{\text{VP}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^T\Sigma^{-1}\mathbf{1}}$$

The primary advantages of Volatility Parity (VP) lie in its simplicity and its independence from asset correlations. As evident in the weight computation, only asset volatilities are considered, while correlations are disregarded. This makes VP particularly useful in situations where estimating the covariance matrix is challenging or unreliable. VP allocates an equal amount of risk (in terms of volatility) to each asset, which should, in principle, ensure a certain level of diversification. However, it

is important to note that the sum of these equal amounts of allocated volatility tends to exceed the total portfolio volatility. This effect arises precisely because correlations are not taken into account. The main drawback of VP is its sensitivity to the characteristics of the asset universe under consideration. To illustrate this, consider two extreme cases: (i) a universe in which all assets exhibit the same volatility, and (ii) a universe where some assets are significantly more volatile than others. In the first case, VP allocation results in an equally weighted portfolio, whereas in the second case, the allocation scheme produces highly concentrated portfolios in the lower-volatility assets. For instance, if the investment universe includes both bonds and equities, VP will naturally lead to a portfolio with a strong overweight in bonds, as they exhibit lower volatility compared to stocks. This may lead to suboptimal risk-adjusted returns under certain market conditions.

Hierarchical Risk Parity Portfolio

Hierarchical Risk Parity (HRP) is a portfolio optimization technique designed to overcome the limitations of mean-variance optimization (MVO), this approach was introduced by de Prado [10]. Traditional MVO, introduced by Markowitz in 1952 [11], assumes that asset returns follow a normal distribution and requires an estimate of the covariance matrix. However, the Modern Portfolio Theory (MPT) model has several drawbacks: forecasting returns is prone to errors, particularly when the number of assets is large; when assets are highly correlated, the covariance matrix becomes ill-conditioned, making inversion unreliable; MVO tends to assign disproportionately high weights to assets with historically high returns, leading to concentrated risk exposure; and the instability of parameter estimates further complicates its application.

These challenges highlight the need for alternative methods that do not rely on return estimation and do not require the invertibility of the covariance matrix. Hierarchical Risk Parity (HRP), introduced in 2016, addresses these issues by leveraging machine learning techniques and graph theory, specifically hierarchical clustering, to build more stable and diversified portfolios. Since HRP does not require the covariance matrix to be invertible, it can effectively compute portfolios even with singular or ill-conditioned covariance matrices. In *out-of-sample* testing, HRP has demonstrated superior performance compared to traditional methods, effectively reducing risk while maintaining returns.

As previously noted, the Critical Line Algorithm (CLA) and other quadratic optimization methods, such as the Markowitz model, guarantee the discovery of the exact solution after a fixed number of iterations, effectively bypassing the Karush-Kuhn-Tucker conditions. However, this does not ensure the accuracy of the solution or provide a well-defined stopping criterion, making CLA solutions potentially unreliable. A significant drawback is that even small deviations in forecasted returns can result in markedly different portfolio allocations.

These limitations have prompted research to focus increasingly on risk budgeting approaches, such as Risk Parity (RP). While these methods enhance portfolio stability, they do not completely eliminate instability. This instability is tied to the condition number of a diagonalizable matrix, such as correlation or covariance matrices, which is defined as the ratio between the largest and smallest eigenvalues. In the special case where the matrix is diagonal with all elements equal to 1, the matrix is its own inverse, and the condition number reaches its minimum. However, when correlations are introduced, the condition number increases, complicating the correlation matrix. Excessively high

condition numbers lead to an unstable inverse matrix, where even minor changes in any element can result in substantial variations in the inverse. This issue is directly relevant to portfolio construction, as the inclusion of highly correlated assets exacerbates instability.

Furthermore, estimation errors—such as those arising from correlations computed using historical data—can ultimately undermine the benefits of diversification. Additionally, to accurately estimate the coefficients of an invertible covariance matrix of dimension N, at least $\frac{N(N+1)}{2}$ independent and identically distributed (IID) observations are required. For large-scale portfolios, this requirement increases significantly. Moreover, it is well-established that correlation structures are not time-invariant.

From a geometric perspective, each entry in the correlation matrix represents the cosine of the angle between the return vectors of assets in the portfolio within the vector space defined by the return series.

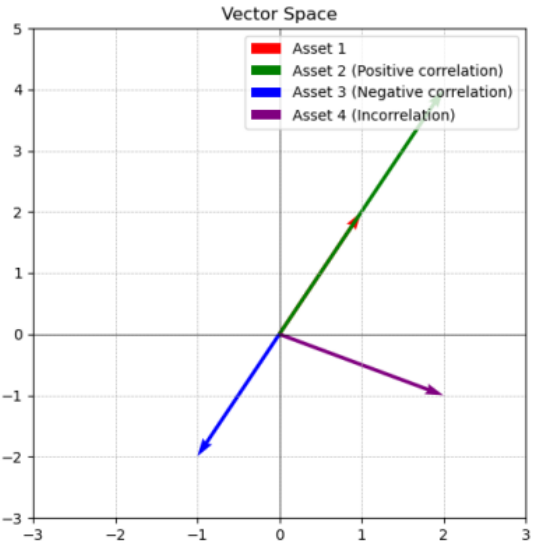


Figure 2: *Correlation matrix from a geometric perspective.*

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos\theta$$

$$\cos\theta = \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}||\mathbf{B}|}$$

$$\cos\theta = \frac{A_x \cdot B_x + A_y \cdot B_y + A_z \cdot B_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}\sqrt{B_x^2 + B_y^2 + B_z^2}}$$

$$\theta = \arccos\left(\frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}||\mathbf{B}|}\right).$$

If the angle is 0°, the correlation, and consequently the cosine, equals 1, indicating that the assets are perfectly correlated and move in unison. When the angle is 90°, the correlation becomes 0, signifying independence between the movements of the assets. Finally, if the angle is 180°, the correlation is -1, suggesting that the assets are perfectly negatively correlated and move in opposite directions.

One of the primary causes of instability in quadratic optimizers is the way the vector space is modeled as a complete graph, where every node is a potential substitute for another. In a correlation matrix, all investments are viewed as potential substitutes for one another, which means that correlation matrices lack any inherent hierarchical structure. This absence of hierarchy allows weights to fluctuate in unintended ways, which is a fundamental source of instability in the Critical Line Algorithm (CLA). This instability is further exacerbated by the complex structure of the correlation matrix, which amplifies even minor estimation errors.

To address this issue, it is essential to introduce a hierarchical structure, typically in the form of a tree. A tree structure, which has only $N-1$ edges to connect N nodes, ensures that portfolio weights are rebalanced only among assets at different hierarchical levels. Furthermore, the distribution of weights follows a top-down approach, leading to more stable portfolio allocations.

The Hierarchical Risk Parity (HRP) algorithm involves three key steps: Tree Clustering, Quasi-Diagonalization, and Recursive Bisection. Each of these steps plays a critical role in ensuring that portfolio weights are allocated in a manner that minimizes risk concentration while maintaining diversification.

Tree clustering Tree Clustering is used to group assets based on their correlation structure, by leveraging a tree-based representation (dendrogram), that methodology enables the identification of clusters of assets exhibiting similar dynamics. The proposed framework consists of several key steps: computation of the correlation matrix, transformation into a distance matrix, hierarchical clustering, and cluster extraction based on predefined linkage criteria.

Compute Pearson correlation matrix Given a return matrix $\mathbf{X}$ of size $T \times N$ (where T is the number of observations and N is the number of assets), compute the Pearson correlation matrix C as:

$$C_{i,j} = \frac{\operatorname{cov}(r_i, r_j)}{\sigma_i \sigma_j}.$$

Where, r_i and r_j are the returns for asset i and j respectively. Whereas, σ_i and σ_j are the standard deviations of the asset i and j respectively.

Distance matrix transformation To convert the correlation matrix into a distance matrix $\mathbf{D}$, it's necessary define a distance measure d: $d : (\mathbf{X}_i, \mathbf{X}_j) \subset B \to \mathbb{R} \in [0, 1]$.

$$d_{i,j} = d[\mathbf{X}_i, \mathbf{X}_j] = \sqrt{\frac{1}{2}(1 - C_{i,j})}.$$

Where B is the Cartesian product of the items in $S = \{1, \ldots, i, \ldots, n\}$.

Distance of Distances: Second-Order Distance Computation A refined distance metric, denoted as $D^*_{i,j}$, is computed based on the Euclidean distance between column vectors of $\mathbf{D}$:

$$D^*_{i,j} = \sqrt{\sum_{n=1}^{N} (D_{n,i} - D_{n,j})^2}.$$

Where $D^*_{i,j}$ encapsulates information from the entire correlation matrix $\mathbf{C}$, since $D_{i,j}$ is defined on column vectors of $\mathbf{X}$, the second-order distance $D^*_{i,j}$, is computed from column vectors of $\mathbf{D}$; thereby capturing higher-order similarity relationships among assets.

Hierarchical Clustering and Linkage Methods Hierarchical clustering is a technique used to construct a tree-based representation of data, enabling the identification of nested groupings without requiring a predefined number of clusters. Within the context of asset allocation, this method facilitates the grouping of assets based on their pairwise dissimilarities, as derived from a distance matrix.

A fundamental step in hierarchical clustering is the selection of a *linkage criterion*, which defines how distances between clusters are computed during the agglomeration process. Among the various linkage methods, **single linkage** is one of the most commonly employed. It merges clusters based on the minimum pairwise distance:

$$(i^*, j^*) = \arg\min_{i,j} D^*_{i,j}$$

where $D^*_{i,j}$ denotes the refined, second-order distance between asset i and asset j, capturing structural similarity across the entire asset universe. When incorporating a new element i, into an existing cluster, the distance is computed as:

$$\min_{j \in \text{cluster}_{n-1}} \left[D^*_{i,j} \right].$$

This approach prioritizes proximity to the closest member within a cluster, often resulting in elongated, chain-like groupings. Other linkage criteria are also used depending on the specific characteristics desired in the clustering structure:

- **Complete linkage**: Merges clusters based on the *maximum* distance between their elements, promoting compact and tightly bound clusters;
- **Ward's method**: Merges clusters to minimize the increase in within-cluster variance, favoring homogeneity and balanced cluster sizes.

The output of hierarchical clustering is a *linkage matrix* $\mathbf{Y} \in \mathbb{R}^{(N-1)\times 4}$, where each row contains information about a cluster merger: the indices of the merged clusters, the distance between them, and the number of original data points contained in the newly formed cluster. This matrix serves as the foundation for constructing a dendrogram, a tree structure that visually represents the hierarchy of merges, which is essential for subsequent portfolio allocation steps in the HRP algorithm.

Quasi-Diagonalization Following the clustering step, the quasi-diagonalization process reorders the covariance matrix to reflect the hierarchical structure revealed by the dendrogram. The primary objective is to position assets with stronger correlations adjacent to each other along the matrix's diagonal. This structural rearrangement enhances numerical stability and interpretability by localizing high covariance values near the main diagonal, thereby concentrating risk within clusters.

To perform this step, the empirical covariance matrix $\mathbf{\Sigma}$ is computed from the asset return matrix $\mathbf{X}$ as:

$$\Sigma_{i,j} = \text{cov}(r_i, r_j),$$

where r_i and r_j represent the return series of assets i and j, respectively. Once calculated, $\mathbf{\Sigma}$ is permuted according to the leaf ordering obtained from hierarchical clustering, yielding a quasi-diagonalized covariance matrix. This structure lays the groundwork for the next stage: recursive allocation of weights.

Recursive Bisection Recursive bisection is the final step in the HRP algorithm, responsible for allocating portfolio weights in a manner that reflects the hierarchical structure previously derived. The method follows a top-down, recursive strategy that distributes capital across sub-clusters in proportion to their estimated risk, specifically their variance.

The procedure begins by assigning equal initial weights to all assets. Then, at each level of the hierarchy, the set of assets is recursively split into two sub-clusters. For each split, the variance of the respective sub-clusters is calculated in matrix form as:

$$\mathbf{V}_i^{(y)} = \mathbf{w}_i^{(j)^T} \mathbf{V}_i^{(j)} \mathbf{w}_i^{(j)}$$

where $\mathbf{V}_i^{(y)}$ is the covariance matrix of cluster j in quadratic form, $\mathbf{V}_i^{(j)}$ is the covariance matrix between the constituents and $\mathbf{w}^{(j)}$ is the inverse-variance weight vector associated with that cluster.

To allocate weights between the two sub-clusters, a split factor β is computed as:

$$\beta = 1 - \frac{\mathbf{V}_i^{(1)}}{\mathbf{V}_i^{(1)} + \mathbf{V}_i^{(2)}}.$$

Cluster 1 receives a weight of β_i, while Cluster 2 receives $(1 - \beta_i)$, ensuring that more capital is allocated to the lower-risk cluster. This procedure is applied recursively down the hierarchy until all terminal nodes (individual assets) receive their final weights. The recursive nature of the algorithm ensures that:

$$0 \leq w_i \leq 1, \quad \forall i \in 1, \ldots, N, \quad \text{and} \quad \sum_{i=1}^{N} w_i = 1.$$

Because weights are partitioned at each level of the hierarchy, the resulting allocation preserves consistency and full investment.

This recursive allocation mechanism contrasts sharply with traditional optimizers like the Critical Line Algorithm (CLA),model the vector space as a complete graph, where each node is treated as a direct substitute for another, leading to erratic weight fluctuations, resulting in unstable weights due to excessive sensitivity to input parameters. HRP, by contrast, imposes structural constraints via the hierarchical tree, reducing the effective degrees of freedom in the optimization and mitigating the amplification of estimation errors.

By incorporating principles of machine learning, risk parity, and classical portfolio theory, HRP provides a robust, interpretable, and computationally efficient alternative to traditional quadratic optimizers. The hierarchical nature of the algorithm ensures that portfolio construction remains stable even in the presence of noisy or ill-conditioned covariance estimates, offering a significant advancement in modern portfolio theory.

3 Data and Methodology

Data description

For this analysis, we consider the sectors introduced in Section 1. Specifically, since one of the ETFs (XLC) was launched in mid-2018, the analysis period for all ETFs has been aligned to this common starting date to ensure consistency and enhance the comparability of results. The time period goes from mid-2018 to the end of May 2025.

Before outlining the methodology, we present a set of summary statistics and diagnostic tests in the following tables. These include the mean, standard deviation, skewness, kurtosis, the Jarque-Bera (JB) test, the Ljung-Box test, and a test for ARCH effects.

The inclusion of these three statistical tests is motivated by their relevance to the assumptions underpinning portfolio modeling. The Jarque-Bera test assesses the normality of return distributions; given that many portfolio models rely on the estimation of distributional moments, significant deviations from normality may lead to biased or misleading parameter estimates (Law of Large Numbers). The Ljung-Box test evaluates the presence of autocorrelation in the return series, which, if significant, could similarly compromise moment estimation and model reliability, as before affecting the Law of Large Numbers assumptions. Lastly, the ARCH test is used to detect conditional heteroskedasticity—i.e., time-varying volatility in the error terms—commonly observed in financial time series, see Campbell at al. [3].

While the outcomes of these tests may have implications for the robustness of parameter estimation and, consequently, for portfolio optimization (e.g., weight allocation), addressing these issues comprehensively lies beyond the scope of the present analysis and will not be pursued further in these notes.

The following results consider the entire sample.

Asset	**Mean (%)**	**Median (%)**	**Std. Dev. (%)**	**Skewness**	**Kurtosis**
XLC	0.0403	0.1133	1.47	-0.48	6.76
XLY	0.0375	0.1267	1.56	-0.54	7.60
XLP	0.0272	0.0557	1.01	-0.47	15.41
XLE	0.0055	0.0800	2.12	-0.96	14.40
XLF	0.0355	0.0837	1.55	-0.57	14.66
XLV	0.0255	0.0548	1.12	-0.44	10.56
XLI	0.0382	0.0875	1.40	-0.51	12.68
XLB	0.0223	0.0587	1.43	-0.42	9.57
XLRE	0.0161	0.0886	1.46	-1.03	16.51
XLK	0.0674	0.1509	1.72	-0.25	8.30
XLU	0.0275	0.0931	1.34	-0.26	14.79

Table 1: *Summary statistics for daily log returns of sector ETFs.*

Asset	JB P-value	LB P-value	ARCH effects P-value
XLC	0.000	0.000	0.000
XLY	0.000	1.40e-14	0.000
XLP	0.000	0.000	0.000
XLE	0.000	1.00e-09	0.000
XLF	0.000	0.000	0.000
XLV	0.000	0.000	0.000
XLI	0.000	0.000	0.000
XLB	0.000	0.000	0.000
XLRE	0.000	0.000	0.000
XLK	0.000	0.000	0.000
XLU	0.000	0.000	0.000

Table 2: *P-values from the Jarque-Bera, Box-Ljung, and ARCH effects tests for the daily log returns of Sector ETFs.*

From Table 1, we observe that, in terms of daily mean returns, XLK achieved the highest performance while XLE recorded the lowest, with all assets generally exhibiting positive returns. Regarding the median, XLK again was the best performer, whereas XLV was the worst. In terms of standard deviation, XLP demonstrated the lowest volatility, and XLE exhibited the highest.

In addition, all assets displayed negative skewness and very high excess kurtosis, suggesting the presence of outliers in the data. These findings on skewness and kurtosis complement the non-normality indicated in Table 2, where all p-values in the Jarque-Bera tests are zero (rejecting the null hypothesis of normality). Moreover, we detected significant autocorrelation in the data; the null hypothesis that the sample autocorrelation coefficients are jointly zero up to lag 10 is rejected, and similar results hold for the ARCH effects (still up to lag 10), as indicated by the LM-type test.

Thus, non-normality, autocorrelation, and heteroscedasticity are all present in the data, which may pose challenges for the standard mean-variance optimization methods employed. However, as previously mentioned, potential adjustments to address these issues are beyond the scope of this analysis.

We can enrich our analysis by examining the correlation matrices among the assets over the full sample period as well as during 2020, a period marked by a pronounced crisis. This approach may provide valuable insights into the varying inter-asset relationships and their potential impacts on Mean-Variance Optimization outcomes.

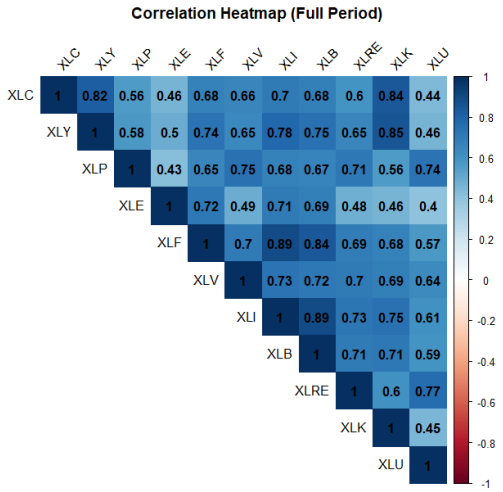


Figure 3: *Correlation matrix full sample.*

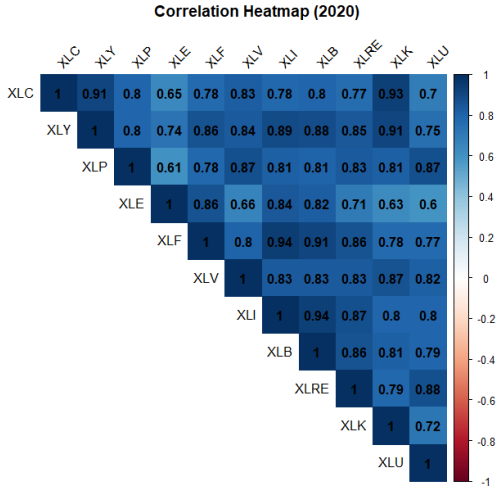


Figure 4: *Correlation matrix 2020 period.*

As observed, all the sectors exhibit positive correlations, which raises concerns about the application of models that typically rely on asset diversification. This issue could adversely affect the reliability of the resulting outcomes. Moreover, the significantly stronger correlations observed during 2020, a period marked by heightened market distress, are particularly critical within the Mean-Variance Optimization framework, potentially leading to unsatisfactory results.

Methodology

Sample split and estimation procedure

To enhance the realism of the analysis and account for potential regime shifts, a moving rebalancing framework is employed. Specifically, two estimation windows are considered—one spanning 12 months and the other 18 months. Rebalancing is conducted at two different frequencies: every months and every 3 months. This results in four distinct configurations: a 12-month window with quarterly

rebalancing, a 12-month window with monthly rebalancing, and the same two rebalancing frequencies applied to the 18-month window.

Each rebalancing point involves rolling the estimation window forward by either 1 or 3 months, depending on the chosen frequency. This structure generates a sequence of *out-of-sample* periods, during which the performance of the portfolio strategy is evaluated.

This methodology better reflects the practices commonly adopted by investment professionals. Moreover, employing a rolling window for the estimation of portfolio weights—specifically, the mean returns and the variance-covariance matrix—may yield more robust results. This is particularly relevant given that portfolio models rely on historical estimates as proxies for future expectations.

Performance evaluation metrics

For this analysis, we employed a variety of risk-adjusted performance metrics. In particular, we considered the Sharpe Ratio, Sortino Ratio, Probabilistic Sharpe Ratio, Jensen's alpha, and two models for assessing market timing: Treynor and Mazuy, and Herniksson and Merton.

Sharpe Ratio The Sharpe Ratio is a measure used in the field of investments to evaluate strategies or portfolios. Introduced in 1966 by the economist William F. Sharpe, [17], this index aims to compare the expected excess return of an investment relative to its volatility. In particular, assuming the risk-free rate equal to zero, our estimated SR is:

$$\widehat{\mathrm{SR}} = \frac{\hat{\mu}}{\hat{\sigma}} \tag{6}$$

Where in (6), μ is generally the expected return (or excess) and σ the standard deviation of our returns (or excess). Both μ and σ are generally estimated via the sample mean and sample standard deviation.

It is possible to contextualize the Sharpe Ratio by treating it as a statistical estimator and conducting inference on it. As demonstrated by Miller and Gher [13], the estimator in equation (6) is closely related to the Student's t-statistic used for testing whether the mean of a random variable is equal to zero. Specifically, the relationship can be expressed as:

$$t = \frac{\hat{\mu}}{\hat{\sigma}/\sqrt{T}} = \sqrt{T}\,\widehat{\mathrm{SR}},$$

where $\hat{\mu}$ is the sample mean of returns, $\hat{\sigma}$ is the sample standard deviation, T is the sample size, and $\widehat{\mathrm{SR}}$ is the estimated Sharpe Ratio.

Hence results that hold for this t-statistic also hold for the Sharpe ratio, up to the scaling factor. So that, the t stat obtained from the t statistics help us in performing hypothesis testing and so assess this value. This formulation allows us to perform statistical inference using the estimated Sharpe Ratio and to assess whether it significantly deviates from a pre-specified benchmark value.

Despite its simplicity, a notable limitation of the Sharpe Ratio lies in its failure to account for autocorrelation within the return series of a given strategy. This issue can be problematic, as emphasized by Lo [9], who demonstrates that in the presence of autocorrelation, the conventional

time aggregation factor (typically expressed as $\sqrt{T}$) may become invalid, thereby rendering inference procedures based on the Sharpe Ratio potentially misleading.

Although adjusting the Sharpe Ratio to account for serial correlation in returns may be beneficial, such corrections are omitted here due to the requirement of explicitly modeling the underlying time series dynamics—e.g., via an autoregressive process of order one, AR(1).

Probabilistic Sharpe ratio The PSR, introduced by Bailey and De Prado [1], is an evolution of the famous Sharpe Ratio. The SR is easy to compute and interpret, however, seen the pros of the SR is also important to underline and understand the limitations of this instrument which lie in its binding assumptions.

The key assumptions underlying the Sharpe Ratio (SR) are the *normality* and *independence and identical distribution (IID)* of asset returns. However, empirical findings show that real-world returns often deviate from the normal distribution, typically exhibiting excess skewness and kurtosis.

Under these assumptions, returns are normally distributed with mean μ and variance σ^2, i.e., $r_t \sim \mathcal{N}(\mu, \sigma^2)$. Since the parameters μ and σ^2 are typically unknown, they must be estimated from historical data. As a result, the Sharpe Ratio itself becomes an estimator, subject to statistical error.

Another limitation of the SR is that it does not characterize a distribution of returns, this means that for any given SR, there can be infinite combinations of normal distributions: the SR is a ratio between two scalars so there are infinite possibilities to reach that value. The fundamental problem with this property is that, regardless from any "good" SR, there might still be the risk of losses, also because the combination of two normal distribution produces infinite couples of skewness and kurtosis. This conclusion can be applied also to the case of non-normality, with the side effect that the confidence interval increases more in case of non-normality. In summary, for any risk adverse investor, if a SR is high, it does not assure an efficient ratio between risk and reward. Also important to notice is that the SR remains unchanged if both expected return and risk increase proportionally. However, a risk adverse investor may do not accept a double return as high enough compensation for a double risk. He may ask for a higher expected return since he typically penalizes risk in a nonlinear way. This inconvenience makes the SR less suitable for capturing risk aversion and risk-taking behavior.

The aim of the PSR is to offer a de inflated metric. Given a benchmark, a fixed Sharpe ratio SR^*, the PSR measure the probability that the estimated Sharpe ratio $\widehat{\mathrm{SR}}$ is higher than SR^*. The formula of the PSR is given by:

$$\widehat{\mathrm{PSR}}(\mathrm{SR}^*) = Z\left[\frac{(\widehat{\mathrm{SR}} - \mathrm{SR}^*)}{\sigma(\widehat{\mathrm{SR}})}\right].$$

Where in the above equation $\sigma(\widehat{\mathrm{SR}}) = \sqrt{\frac{1-\hat{\gamma}_3\widehat{SR}+\frac{\hat{\gamma}_4-1}{4}\widehat{SR}^2}{N-1}}$.

Where Z is the cumulative distribution function of the standard Normal distribution, $\hat{\gamma}_3$ is the estimated skewness, $\hat{\gamma}_4$ is the estimated excess kurtosis while N indicates the number of observations. From the formula it is easy to notice that PSR increases, for a given SR^*, for a greater $\widehat{\mathrm{SR}}$ or for a bigger number of observations. The skewness and kurtosis impact respectively with positive and negative impact: positive skewness and fatter tails. To sum up the elemetns that the PSR take into account, are: track record's non-normality, length and sampling frequency. Moreover, another

important aspect of the PSR is that since hedge fund strategies are usually characterized by negative skewness and fat tails, Sharpe ratios tend to be "inflated". The PSR, takes those characteristics into account and delivers a corrected, atemporal, (no annualization performed) measure of performance expressed in terms of probability of skill.

We recall that, the PSR gives you the probability of actually being superior with respect to a SR benchmark denoted as SR^*. This probability should be compared with the confidence level considered, for instance 95% or 99%.

Sortino Ratio The Sortino Ratio, introduced by Frank Sortino, constitutes a refinement of the Sharpe Ratio and stands as one of the most appreciated metrics for evaluating return relative to risk. It is particularly employed for comparative analysis among portfolios or investment funds and was developed to address critical limitations inherent in the use of standard deviation, as adopted in the Sharpe Ratio. These limitations include:

- The assumption of symmetric return distributions, a premise frequently challenged as unrealistic in empirical finance;
- The equal penalization of positive and negative deviations from the mean;
- The reliance on the mean return as the benchmark for evaluating performance.

To overcome these shortcomings, the Sortino Ratio measures the excess return of an investment or strategy relative to the *Minimum Acceptable Return (MAR)*, incorporating solely the Downside Deviation in its denominator:

$$\text{Sortino Ratio} = \frac{R_p - \text{MAR}}{\sigma_d}.$$

Here, the MAR represents the minimum return threshold deemed acceptable by the investor for a particular investment. This parameter, which may be selected subjectively or determined with reference to a market benchmark, delineates the boundary of perceived risk: returns falling below this level are considered undesirable.

The Downside Deviation, in contrast to the conventional standard deviation, replaces the mean return with the MAR and accounts exclusively for observations that fall short of this threshold. This formulation reflects the practical investor perspective that only returns beneath the acceptable level are genuinely perceived as risky.

This measure of risk is defined as the square root of the Downside Variance, computed as the mean of squared deviations of observed returns beneath the MAR. Formally, the Downside Deviation is expressed as:

$$\sigma_d = \sqrt{\frac{\sum_{i=1}^{N} \min[(R_i - \text{MAR}), 0]^2}{N}}.$$

Jensen's Alpha The Capital Asset Pricing Model (CAPM), introduced by Sharpe [18], builds upon Markowitz's portfolio theory and establishes a theoretical framework linking the expected return of a financial asset to its associated risk. Initially developed to explain asset pricing under market

equilibrium, the CAPM was later recognized for its practical utility in enabling investors to construct optimal portfolios by combining a risk-free asset with the market portfolio.

In accordance with CAPM theory, we define an index intended to evaluate the performance of an investment fund while accounting for its risk exposure. Performance assessment involves more than merely analyzing the fund's raw returns; it requires comparison with an appropriate, rational CAPM-based strategy. Typically, the market portfolio of the CAPM represents the fund's benchmark, i.e., the market index the fund purports to replicate or outperform.

The CAPM is formally represented by the following equation:

$$\mathbb{E}[r_p] = r_f + \beta_{p,m}(\mathbb{E}[r_m] - r_f).$$

This equation articulates the relationship between the expected return of a portfolio, $\mathbb{E}[r_p]$, and that of the market, $\mathbb{E}[r_m]$, where r_f denotes the risk-free rate, and $\beta_{p,m}$ is the sensitivity of the portfolio to market movements.

To empirically estimate the expected return expressed above, one can perform a linear regression in which the portfolio's excess returns serve as the response variable, and the market's excess returns (i.e., market return minus the risk-free rate) act as the regressor. The associated assumed data-generating process is given by:

$$r_{t,p} - r_{t,f} = \alpha + \beta_{p,m}(r_{t,m} - r_{t,f}) + \epsilon_t.$$

Where, $r_{t,p}$ is the portfolio return at time t, and $r_{t,m}$ is the market portfolio return at time t. Our primary interest lies in the parameter α, commonly referred to as Jensen's alpha. This coefficient quantifies the extent to which a portfolio outperforms the return predicted by the CAPM, thus capturing abnormal performance. A positive alpha indicates superior returns beyond those justified by the market risk, signifying an effective investment strategy. For further elaboration, see Jensen [7].

Treynor and Mazuy The Treynor and Mazuy model [15], is employed to assess market timing ability. This specification extends the basic CAPM by incorporating a quadratic term to capture non-linearity in the relationship between market and portfolio returns. It is defined as follows:

$$r_{t,p} - r_{t,f} = \alpha + \beta_{p,m}(r_{t,m} - r_{t,f}) + \gamma_i(r_{t,m} - r_{t,f})^2 + \epsilon_t.$$

The parameter γ serves as a measure of market timing ability. We anticipate that a statistically significant, positive value of γ indicates effective market timing, thereby suggesting *macroforecasting* competence.

Henriksson and Merton The Henriksson and Merton model [6], provides an alternative extension of the CAPM by conceptualizing market timing as akin to offering a free put option to investors, thereby protecting them during market downturns. Its formulation is given by:

$$r_{t,p} - r_{t,f} = \alpha + \beta_{p,m}(r_{t,m} - r_{t,f}) + \gamma_i D(r_{t,m} - r_{t,f}) + \epsilon_t.$$

where the dummy variable D is defined as:

$$D = \begin{cases} 0, & \text{if} \quad r_{t,m} - r_{t,f} \geq 0 \\ -1, & \text{if} \quad r_{t,m} - r_{t,f} < 0. \end{cases}$$

In this context, the parameter γ again plays a critical role. A positive and significant estimate for γ would provide evidence of market timing ability, reinforcing the notion of an inherent *macroforecasting* capability within the portfolio strategy.

Block Bootstrap For all the metrics considered in this analysis, we applied a block bootstrap procedure to account for the temporal dependencies inherent in our data. Traditional bootstrap methods, as introduced by Efron [5], assume that observations are independent and identically distributed (i.i.d.), an assumption that does not hold for time series data. Block bootstrap techniques overcome this limitation by preserving the dependence structure within the data.

The central idea of block bootstrapping is to partition the time series into contiguous blocks of observations and then resample these blocks with replacement to construct new pseudo-time series. This approach maintains the local dependence within each block while allowing for variability across the resampled series. One of the earliest and most widely adopted techniques is the Moving Block Bootstrap (MBB), as proposed by Künsch [8], which involves creating overlapping blocks of a fixed length. Alternatively, the Stationary Bootstrap, introduced by Politis and Romano [14], selects blocks of random lengths to better reflect the stationary properties of the series.

Selecting an appropriate block length is crucial: if the blocks are too short, the method will fail to capture the full dependence structure of the data; if they are too long, the variability of the estimators may increase significantly.

4 Empirical Results

We now present the portfolio results based on the evaluation metrics detailed in the previous section, namely the Sharpe ratio, Sortino ratio, Probabilistic Sharpe ratio, Jensen's alpha, Treynor-Mazuy model, and the Henriksson and Merton model.

It is important to note that we employed a moving window approach for the estimation of portfolio weights. Specifically, we considered rolling windows of 12 and 18 months, coupled with two distinct rebalancing frequencies: monthly and quarterly. This methodology results in four different configurations of outcomes, each reflecting the impact of window length and rebalancing periodicity on the performance metrics. Only the *out-of-sample* results of each portfolio were analyzed; that is, we focused on the performance during the periods occurring between rebalancing dates.

Given the relatively limited quantity of data available, we utilized a moving block bootstrapping procedure to enhance the precision of the estimates for the Sharpe, Sortino, and Probabilistic Sharpe ratios. The block length was chosen to be 3 in order to account for potential autocorrelations within the data. We acknowledge, however, that the selection of an appropriate block length should ideally be based on a thorough analysis of the autocorrelation function (ACF) or through the implementation of formal time series models, such as those within an autoregressive framework. In this instance, the block size was selected arbitrarily without the benefit of a formal test.

We begin by examining the plots that illustrate the cumulative returns across the different portfolios, taking into account various window sizes and rebalancing frequencies.

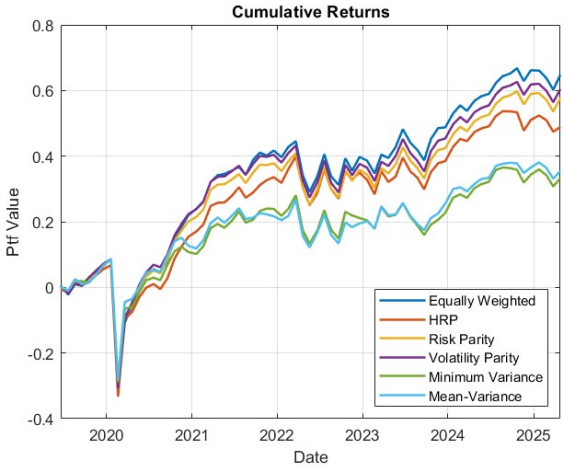


Figure 5: *Cumulative returns for one-year window and monthly rebalancing.*

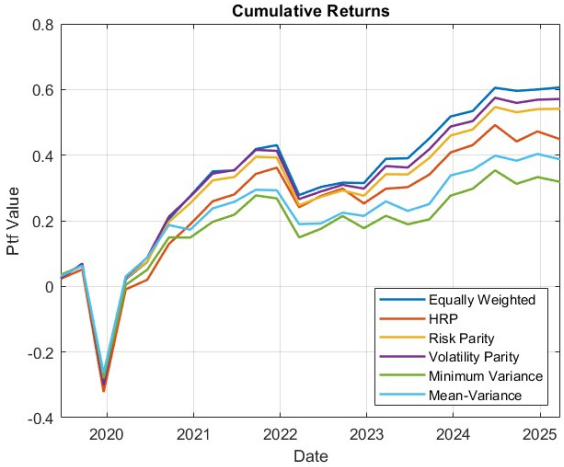


Figure 6: *Cumulative returns for one-year window and quarterly rebalancing.*

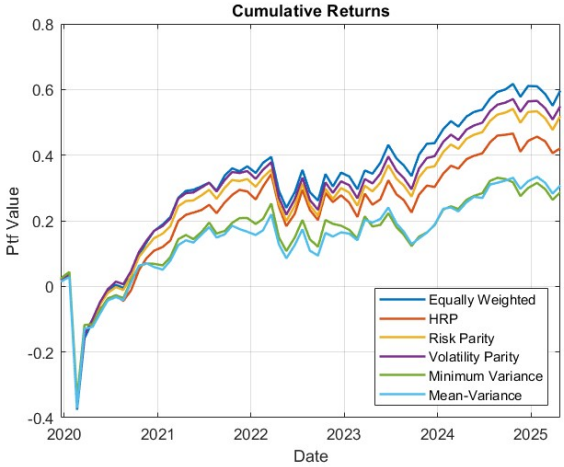


Figure 7: *Cumulative returns for 18-months window and monthly rebalancing.*

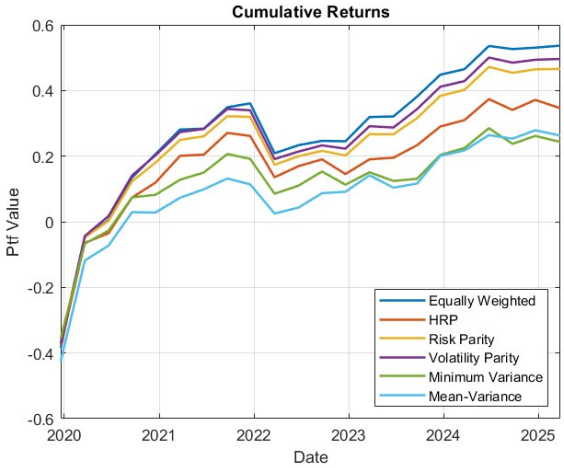


Figure 8: *Cumulative returns for 18-month window and quarterly rebalancing.*

From the figures above, we observe that across varying window sizes and rebalancing frequencies, all portfolios exhibit remarkably similar performance, indicating minimal differences. When considering the different models, a ranking based on the past three years of performance yields the following order—from best to worst: Equally Weighted, Volatility Parity, Risk Parity, HRP, Mean-Variance, and Minimum Variance. These results clearly underscore the challenges associated with constructing portfolios using traditional methods. The following paragraph further elaborates on the return profiles by incorporating risk-adjusted measures.

Results for the Sharpe, Sortino, and Probabilistic Sharpe Ratios

The following tables present the results for the mean Sharpe, Sortino, and Probabilistic Sharpe ratios across all configurations of window lengths and rebalancing frequencies, obtained from moving block bootstrap. The number of bootstrap replications is equal to 10,000, the mean, 20%, and 80% percentile were considered in the following Tables. We consider the annualized Sharpe and Sortino, moreover, a risk-free of zero and a $SR^* = 0$ is assumed.

Portfolio	**Sharpe Ratio**	**Sortino Ratio**	**PSR**
Equally Weighted	0.588	1.000	85.4
	(0.304, 0.907)	(0.393, 1.657)	(75.0, 98.5)
HRP	0.445	0.763	80.4
	(0.197, 0.706)	(0.253, 1.272)	(67.4, 96.0)
Risk Parity	0.534	0.911	83.7
	(0.262, 0.834)	(0.341, 1.521)	(72.3, 97.9)
Volatility Parity	0.556	0.944	84.4
	(0.280, 0.863)	(0.359, 1.585)	(73.3, 98.2)
Minimum Variance	0.334	0.579	74.6
	(0.102, 0.573)	(0.131, 1.005)	(59.5, 92.2)
Mean-Variance	0.356	0.618	75.8
	(0.119, 0.597)	(0.153, 1.043)	(61.0, 92.9)

Table 3: *Block bootstrapped estimates for the Sharpe, Sortino, and Probabilistic Sharpe ratios. A rolling one-year window with monthly rebalancing is employed, and values in parentheses indicate the 20th and 80th percentile estimates.*

Portfolio	**Sharpe Ratio**	**Sortino Ratio**	**PSR**
Equally Weighted	0.467	0.997	80.577
	(0.205, 0.769)	(0.279, 1.617)	(68.222, 96.265)
HRP	0.355	0.721	75.590
	(0.125, 0.614)	(0.165, 1.183)	(61.599, 92.671)
Risk Parity	0.434	0.922	79.247
	(0.183, 0.727)	(0.247, 1.503)	(66.490, 95.539)
Volatility Parity	0.453	0.964	80.086
	(0.198, 0.748)	(0.267, 1.563)	(67.549, 95.876)
Minimum Variance	0.247	0.506	68.751
	(0.025, 0.471)	(0.032, 0.845)	(52.460, 86.963)
Mean-Variance	0.346	0.737	74.536
	(0.101, 0.615)	(0.134, 1.207)	(59.554, 92.918)

Table 4: *Block bootstrapped estimates for the Sharpe, Sortino, and Probabilistic Sharpe ratios. A rolling one-year window with quarterly rebalancing is employed, and values in parentheses indicate the 20th and 80th percentile estimates.*

Portfolio	**Sharpe Ratio**	**Sortino Ratio**	**PSR**
Equally Weighted	0.517	0.877	81.907
	(0.239, 0.837)	(0.304, 1.507)	(69.735, 97.313)
HRP	0.350	0.597	74.859
	(0.120, 0.597)	(0.152, 1.024)	(60.581, 91.981)
Risk Parity	0.449	0.763	79.362
	(0.190, 0.739)	(0.240, 1.307)	(66.115, 95.706)
Volatility Parity	0.479	0.817	80.595
	(0.211, 0.774)	(0.270, 1.387)	(67.792, 96.440)
Minimum Variance	0.245	0.433	68.507
	(0.027, 0.469)	(0.035, 0.789)	(52.514, 86.413)
Mean-Variance	0.250	0.437	68.743
	(0.031, 0.479)	(0.040, 0.789)	(52.892, 86.831)

Table 5: *Block bootstrapped estimates for the Sharpe, Sortino, and Probabilistic Sharpe ratios. A rolling one-year and half window with monthly rebalancing is employed, and values in parentheses indicate the 20th and 80th percentile estimates.*

Portfolio	**Sharpe Ratio**	**Sortino Ratio**	**PSR**
Equally Weighted	0.568	1.169	88.387
	(0.402, 0.723)	(0.680, 1.570)	(82.139, 95.499)
HRP	0.385	0.765	80.086
	(0.235, 0.530)	(0.370, 1.096)	(70.760, 90.287)
Risk Parity	0.517	1.058	86.471
	(0.356, 0.670)	(0.585, 1.443)	(79.303, 94.501)
Volatility Parity	0.544	1.132	87.613
	(0.380, 0.696)	(0.636, 1.541)	(80.833, 95.226)
Minimum Variance	0.302	0.614	74.505
	(0.139, 0.466)	(0.211, 0.972)	(62.761, 87.857)
Mean-Variance	0.298	0.582	73.849
	(0.139, 0.455)	(0.201, 0.922)	(62.592, 87.193)

Table 6: *Block bootstrapped estimates for the Sharpe, Sortino, and Probabilistic Sharpe ratios. A rolling one-year and half window with quarterly rebalancing is employed, and values in parentheses indicate the 20th and 80th percentile estimates.*

From Table 3, we observe that the performance metrics are not particularly attractive, as evidenced by the low annualized Sharpe ratios. In contrast, the Sortino ratios are higher, suggesting that the portfolios exhibit better exposure to upside risk relative to downside risk. Notably, even in the top 20% scenarios, both the Sharpe and Sortino ratios remain modest, especially for the Minimum Variance and Mean Variance approaches. Moreover, the PSR results indicate that none of the strategies demonstrate skill beyond the no-skill benchmark ($SR^* = 0$); in fact, none of the estimates exceed the conventional confidence intervals of 90%, 95%, or 99%.

From Table 4, we note that both the Sharpe and Sortino ratios are slightly lower than those obtained with monthly rebalancing. The PSR values follow a worst pattern, remaining below the typical thresholds defined by the 90%, 95%, and 99% confidence levels.

When considering an 18-month rolling window, as shown in Table 5, the performance appears to deteriorate slightly across all evaluation metrics compared to Table 3. Finally, Table 6 presents arguably the most favorable results in terms of the Sortino ratio and PSR when compared with other window size and rebalancing configurations.

If one were to rank the different portfolio strategies from best to worst, the order would be: EW, Volatility Parity, Risk Parity, HRP, Mean Variance, and Minimum Variance.

In the following section, we analyze the estimated α obtained from the regressions against the S&P 500 benchmark.

Alpha Estimation Results

The results presented are based on the following regression:

$$r_{t,p} = \alpha + \beta\, r_{t,m} + \epsilon_t,$$

where, for simplicity, we assume that $r_f = 0$ (risk free rate), and that r_m is the S&P 500. We acknowledge that excluding the risk-free rate may have a significant impact, especially over the time period under analysis.

Our primary objective is to examine the estimated intercept, $\hat{\alpha}$, which represents the excess return, the goal being to obtain $\hat{\alpha} > 0$. Furthermore, to address issues arising from autocorrelation and heteroscedasticity, HAC standard errors were employed for computing the standard errors and p-values, thereby facilitating an accurate assessment of statistical significance. The following tables present the results for all configurations of window sizes and rebalancing frequencies.

Portfolio	**Alpha**	**P-value**
Equally Weighted	-0.002	0.166
HRP	-0.004	0.114
Risk Parity	-0.003	**0.077**
Volatility Parity	-0.003	**0.097**
Minimum Variance	-0.006	**0.013**
Mean-Variance	-0.005	**0.016**

Table 7: *Estimated $\hat{\alpha}$ and its associated p-value computed using HAC standard errors. A one-year rolling window with monthly rebalancing was employed.*

Portfolio	**Alpha**	**P-value**
Equally Weighted	-0.006	0.361
HRP	-0.010	0.312
Risk Parity	-0.007	0.263
Volatility Parity	-0.006	0.289
Minimum Variance	-0.014	0.106
Mean-Variance	-0.010	0.182

Table 8: *Estimated $\hat{\alpha}$ and its associated p-value computed using HAC standard errors. A one-year rolling window with quarterly rebalancing was employed.*

Portfolio	**Alpha**	**P-value**
Equally Weighted	-0.002	0.186
HRP	-0.005	**0.057**
Risk Parity	-0.003	**0.068**
Volatility Parity	-0.003	**0.092**
Minimum Variance	-0.006	**0.021**
Mean-Variance	-0.006	**0.034**

Table 9: *Estimated $\hat{\alpha}$ and its associated p-value computed using HAC standard errors. A 18-month rolling window with monthly rebalancing was employed.*

Portfolio	**Alpha**	**P-value**
Equally Weighted	-0.006	0.390
HRP	-0.012	0.186
Risk Parity	-0.008	0.249
Volatility Parity	-0.007	0.274
Minimum Variance	-0.014	0.140
Mean-Variance	-0.017	0.185

Table 10: *Estimated $\hat{\alpha}$ and its associated p-value computed using HAC standard errors. A 18-month rolling window with quarterly rebalancing was employed.*

From the tables above, it is apparent that none of the strategies generate positive abnormal returns. Moreover, several of the associated p-values are not statistically significant (only those highlighted in bold meet the significance threshold of $\alpha = 10\%$, and only those within the montly rebalancing are). Even though the EW portfolio tends to exhibit the best performance metrics (in the previous paragraph), it still produces an insignificant negative α, a finding that holds true for the other portfolios, despite their relatively strong Sortino, Sharpe, and PSR values.

Results of Market Timing Models

In this paragraph, we present the results from the market timing models explained in the previous section. More specifically, we present the results for both models. The regression models considered are the following:

$$\text{Treynor Mazuy} \quad r_{t,p} = \alpha + \beta_{p,m}(r_{t,m}) + \gamma_i(r_{t,m})^2 + \epsilon_t.$$

$$\text{Henriksson Merton} \quad r_{t,p} = \alpha + \beta_{p,m}(r_{t,m}) + \gamma_i D(r_{t,m}) + \epsilon_t.$$

For both models, the risk-free rate is set to zero. Our primary objective is to assess the sign and significance of the parameter γ. Consequently, we focus solely on reporting the estimated $\hat{\alpha}$ and $\hat{\gamma}$ parameters, where $\hat{\gamma}$ pertains to the quadratic term in one model and to the dummy variable in the other. Additionally, HAC (Heteroskedasticity and Autocorrelation Consistent) robust standard errors have been employed in both models.

Portfolio	$\hat{\alpha}$		**Dummy Coefficient**	
	Estimate	**P-value**	**Estimate**	**P-value**
Equally Weighted	-0.007	0.002	-0.010	0.035
HRP	-0.011	0.000	-0.018	0.013
Risk Parity	-0.008	0.000	-0.012	0.023
Volatility Parity	-0.007	0.000	-0.010	0.040
Minimum Variance	-0.015	0.000	-0.023	0.001
Mean-Variance	-0.015	0.000	-0.024	0.000

Table 11: *Henriksson and Merton model. Estimated $\hat{\alpha}$ and dummy variable coefficient along with their p-values. The results refer to the one-year window and monthly rebalancing. HAC standard errors considered.*

Portfolio	$\hat{\alpha}$		**Squared Coefficient**	
	Estimate	**P-value**	**Estimate**	**P-value**
Equally Weighted	-0.001	0.538	-0.220	0.077
HRP	-0.003	0.347	-0.194	0.407
Risk Parity	-0.002	0.311	-0.174	0.235
Volatility Parity	-0.002	0.343	-0.169	0.175
Minimum Variance	-0.004	0.134	-0.196	0.448
Mean-Variance	-0.004	0.150	-0.134	0.648

Table 12: *Treynor and Mazuy model. Estimated $\hat{\alpha}$ and squared variable coefficient along with their p-values. The results refer to the one-year window and monthly rebalancing. HAC standard errors considered.*

Portfolio	$\hat{\alpha}$		**Dummy Coefficient**	
	Estimate	**P-value**	**Estimate**	**P-value**
Equally Weighted	-0.020	0.007	-0.049	0.004
HRP	-0.030	0.012	-0.069	0.004
Risk Parity	-0.019	0.014	-0.044	0.011
Volatility Parity	-0.019	0.016	-0.043	0.015
Minimum Variance	-0.025	0.031	-0.040	0.110
Mean-Variance	-0.022	0.025	-0.041	0.105

Table 13: *Henriksson and Merton model. Estimated $\hat{\alpha}$ and dummy variable coefficient along with their p-values. The results refer to the one-year window and quarterly rebalancing. HAC standard errors considered.*

Portfolio	$\hat{\alpha}$		**Squared Coefficient**	
	Estimate	**P-value**	**Estimate**	**P-value**
Equally Weighted	-0.004	0.506	-0.131	0.118
HRP	-0.008	0.443	-0.148	0.265
Risk Parity	-0.005	0.375	-0.112	0.188
Volatility Parity	-0.005	0.372	-0.091	0.237
Minimum Variance	-0.012	0.180	-0.125	0.329
Mean-Variance	-0.010	0.212	-0.027	0.831

Table 14: *Treynor and Mazuy model. Estimated $\hat{\alpha}$ and squared variable coefficient along with their p-values. The results refer to the one-year window and quarterly rebalancing. HAC standard errors considered.*

Portfolio	$\hat{\alpha}$		**Dummy Coefficient**	
	Estimate	**P-value**	**Estimate**	**P-value**
Equally Weighted	-0.007	0.003	-0.011	0.047
HRP	-0.011	0.000	-0.016	0.036
Risk Parity	-0.008	0.000	-0.012	0.041
Volatility Parity	-0.007	0.000	-0.010	0.052
Minimum Variance	-0.016	0.000	-0.025	0.001
Mean-Variance	-0.016	0.000	-0.025	0.001

Table 15: *Henriksson and Merton model. Estimated $\hat{\alpha}$ and dummy variable coefficient along with their p-values. The results refer to the 18-month window and monthly rebalancing. HAC standard errors considered.*

Portfolio	$\hat{\alpha}$		**Squared Coefficient**	
	Estimate	**P-value**	**Estimate**	**P-value**
Equally Weighted	-0.001	0.563	-0.213	0.090
HRP	-0.004	0.229	-0.150	0.499
Risk Parity	-0.002	0.278	-0.168	0.250
Volatility Parity	-0.002	0.327	-0.167	0.179
Minimum Variance	-0.005	0.172	-0.256	0.341
Mean-Variance	-0.005	0.214	-0.281	0.378

Table 16: *Treynor and Mazuy model. Estimated $\hat{\alpha}$ and squared variable coefficient along with their p-values. The results refer to the 18-month window and monthly rebalancing. HAC standard errors considered.*

Portfolio	$\hat{\alpha}$		**Dummy Coefficient**	
	Estimate	**P-value**	**Estimate**	**P-value**
Equally Weighted	-0.021	0.007	-0.051	0.004
HRP	-0.032	0.011	-0.064	0.014
Risk Parity	-0.022	0.011	-0.048	0.009
Volatility Parity	-0.020	0.015	-0.045	0.014
Minimum Variance	-0.029	0.026	-0.050	0.075
Mean-Variance	-0.031	0.029	-0.049	0.127

Table 17: Henriksson and Merton model. *Estimated $\hat{\alpha}$ and dummy variable coefficient along with their p-values. The results refer to the 18-month window and quarterly rebalancing. HAC standard errors considered.*

Portfolio	$\hat{\alpha}$		**Squared Coefficient**	
	Estimate	**P-value**	**Estimate**	**P-value**
Equally Weighted	-0.004	0.542	-0.131	0.101
HRP	-0.010	0.268	-0.135	0.268
Risk Parity	-0.006	0.361	-0.115	0.153
Volatility Parity	-0.005	0.366	-0.098	0.169
Minimum Variance	-0.013	0.199	-0.088	0.508
Mean-Variance	-0.012	0.317	-0.311	0.116

Table 18: *Treynor and Mazuy model. Estimated $\hat{\alpha}$ and squared variable coefficient along with their p-values. The results refer to the 18-month window and quarterly rebalancing. HAC standard errors considered.*

These subpar results, derived from both the alpha estimation model and the market timing ability model relative to the S&P 500, underscore the inherent challenge of generating a sustainable market edge. The literature consistently highlights that achieving consistent outperformance is exceptionally

difficult, even for managers with advanced strategies.

Our study, which examines the period from 2018 to April 2025, further reinforces these findings. This interval has been especially turbulent, marked by the profound disruptions of the COVID-19 pandemic and further market disturbances beginning in 2022.

Conclusions

The results obtained clearly demonstrate the inherent difficulties of generating *alpha* and devising strategies that consistently outperform the market. These challenges stem from several factors. First of all, despite diversifying across various sectors to represent the global economy, the choice of assets itself poses limitations. Moreover, the analysis period—from 2018 to April 2025—has been fraught with uncertainty and crisis, further complicating the performance of even the most sophisticated strategies.

In periods of significant turbulence and uncertainty, classical frameworks such as Mean-Variance Optimization (MVO) encounter substantial challenges. A critical assumption underlying MVO is that asset return distributions are approximately normal, or at least not markedly non-normal. However, during high-volatility periods, distributions frequently exhibit elevated kurtosis. This increased kurtosis undermines the stability of the first two moments of the distribution, as it slows the convergence of the Law of Large Numbers, thereby reducing the precision in estimating the mean and variance. Consequently, the fundamental statistical assumptions of MVO are compromised, which can lead to highly suboptimal results.

Furthermore, as highlighted in the data description section, asset correlations tend to intensify during turbulent periods. Higher correlations erode the benefits of diversification—a cornerstone of the MVO framework—resulting in optimized portfolios that often resemble those produced by an equally weighted approach. In such environments, not only does MVO falter, but alternative models such as volatility parity and risk parity are also adversely affected due to the diminished effectiveness of diversification.

Our examination also extended to the Hierarchical Risk Parity (HRP) approach, but this alternative similarly produced unsatisfactory outcomes during turbulent periods. Paradoxically, the equally weighted (EW) portfolio, tended to perform better than more complex models under these challenging market conditions. A possible reason related to these results might be the high correlation observed.

In general, the weight estimation relies on the hypothesis that the historical distribution computed over a chosen window is sufficiently representative for future projections. However, this assumption is highly debatable.

One might then ask whether forecasting the future distribution could yield better results. A promising alternative involves the use of models such as GARCH or DCC, which are specifically designed to capture time-varying volatility and correlations. Additionally, incorporating regime switching models can help account for abrupt shifts in market conditions.

Nevertheless, these alternative approaches introduce significant additional complexities. In particular, the forecasting horizon, often tied to the rebalancing frequency, which may be quite extended, complicates matters, as do the underlying assumptions regarding volatility dynamics and regime transitions.

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